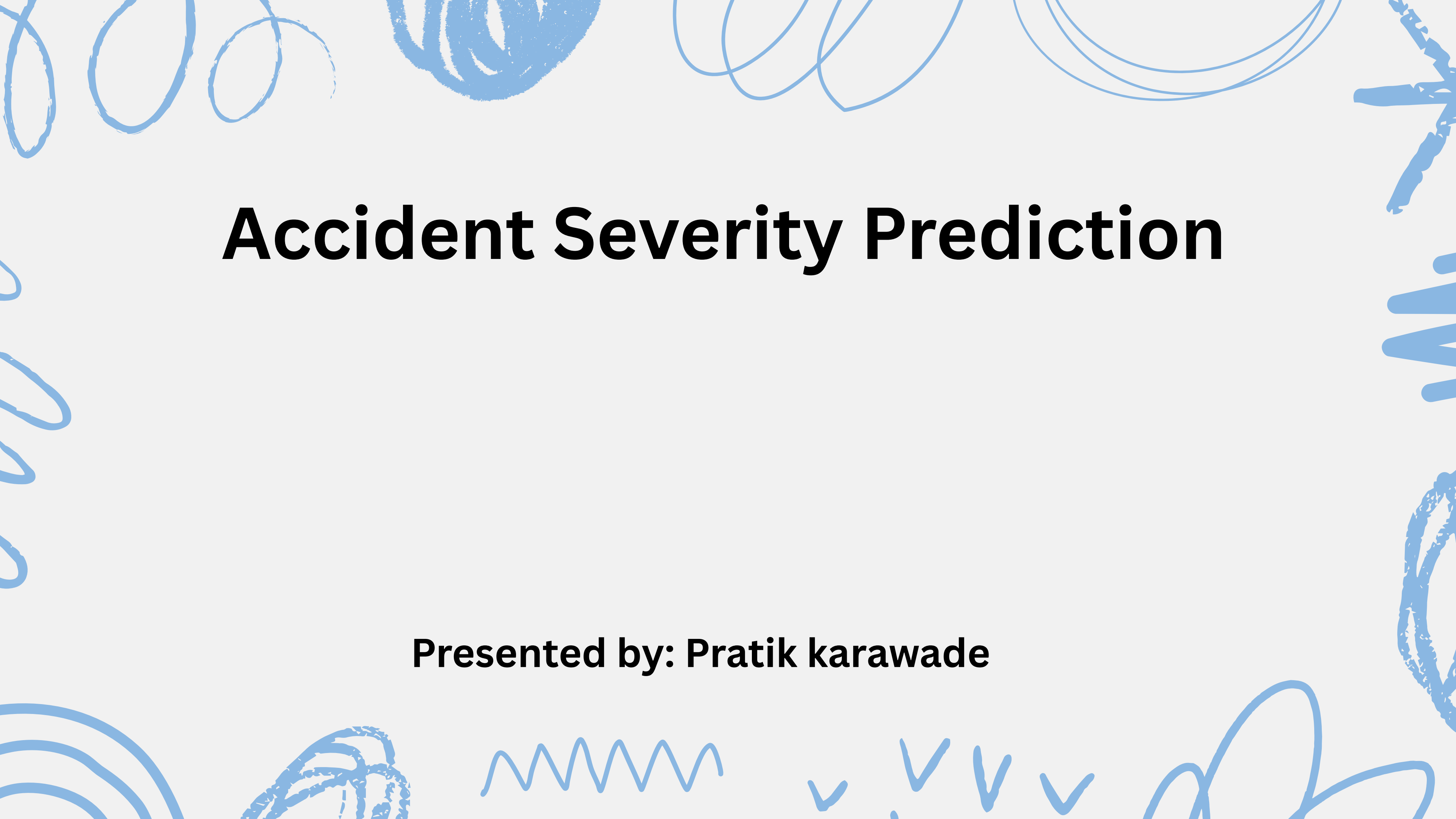


The background of the slide is white and decorated with various blue hand-drawn doodles. These include several overlapping circles and loops at the top, a series of horizontal wavy lines at the bottom, and some abstract scribbles on the right side. The main title is centered in a large, bold, black font.

Accident Severity Prediction

Presented by: Pratik karawade

Goal

- **Develop a machine learning model to predict the severity of car accidents.**
- **Use real-world data to assist emergency responders and support safety initiatives.**

Why It Matters

- **Severity prediction helps save lives, guide safety efforts, and improve emergency response.**

Real-World Benefits

1. Saving Lives

- Enables emergency teams to prioritize severe cases faster.

2. Data-Driven Safety

- Identifies causes of severe accidents (e.g., speeding, weather, road conditions).

3. Resource Optimization

- Allocates personnel and resources more efficiently in real time.

Model Development

Algorithms Explored

- Logistic Regression
- Random Forest
- XGBoost (Final Model)

Model Tuning

- Grid search for hyperparameters
- Cross-validation for stability

IMPORT LIBRARIES AND HANDLE MISSING VALUE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, roc_auc_score, roc_curve, acc
from xgboost import XGBClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
import shap

df = pd.read_csv("/content/Vehicle Collision.csv")
```

df.isnull().sum()	
	0
UNIQUE_ID	0
COLLISION_ID	4
CRASH_DATE	4
CRASH_TIME	4
PERSON_ID	7
PERSON_TYPE	4
PERSON_INJURY	4
VEHICLE_ID	48723
PERSON_AGE	997
EJECTION	127873
EMOTIONAL_STATUS	103742
BODILY_INJURY	103701
POSITION_IN_VEHICLE	127754

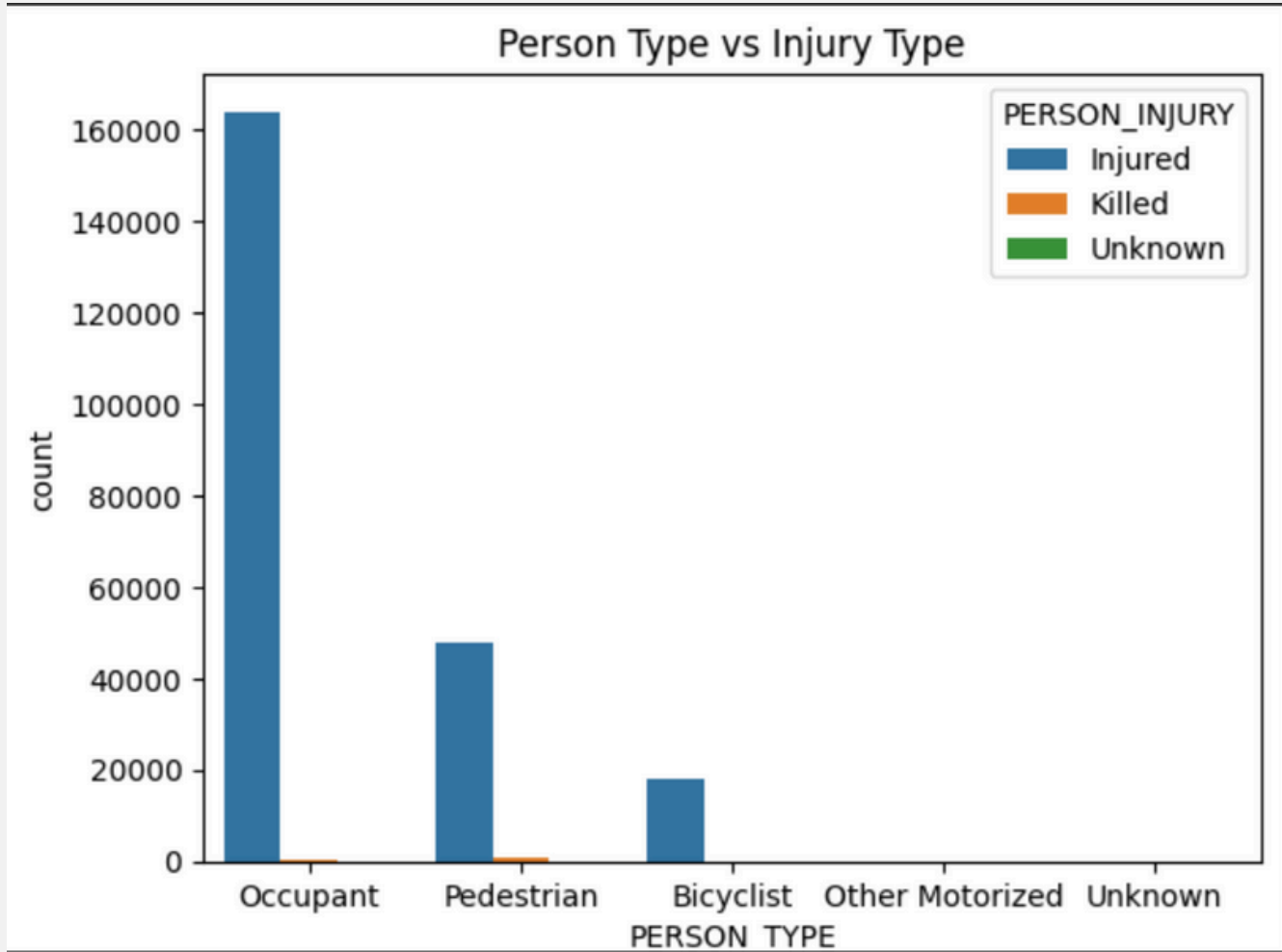
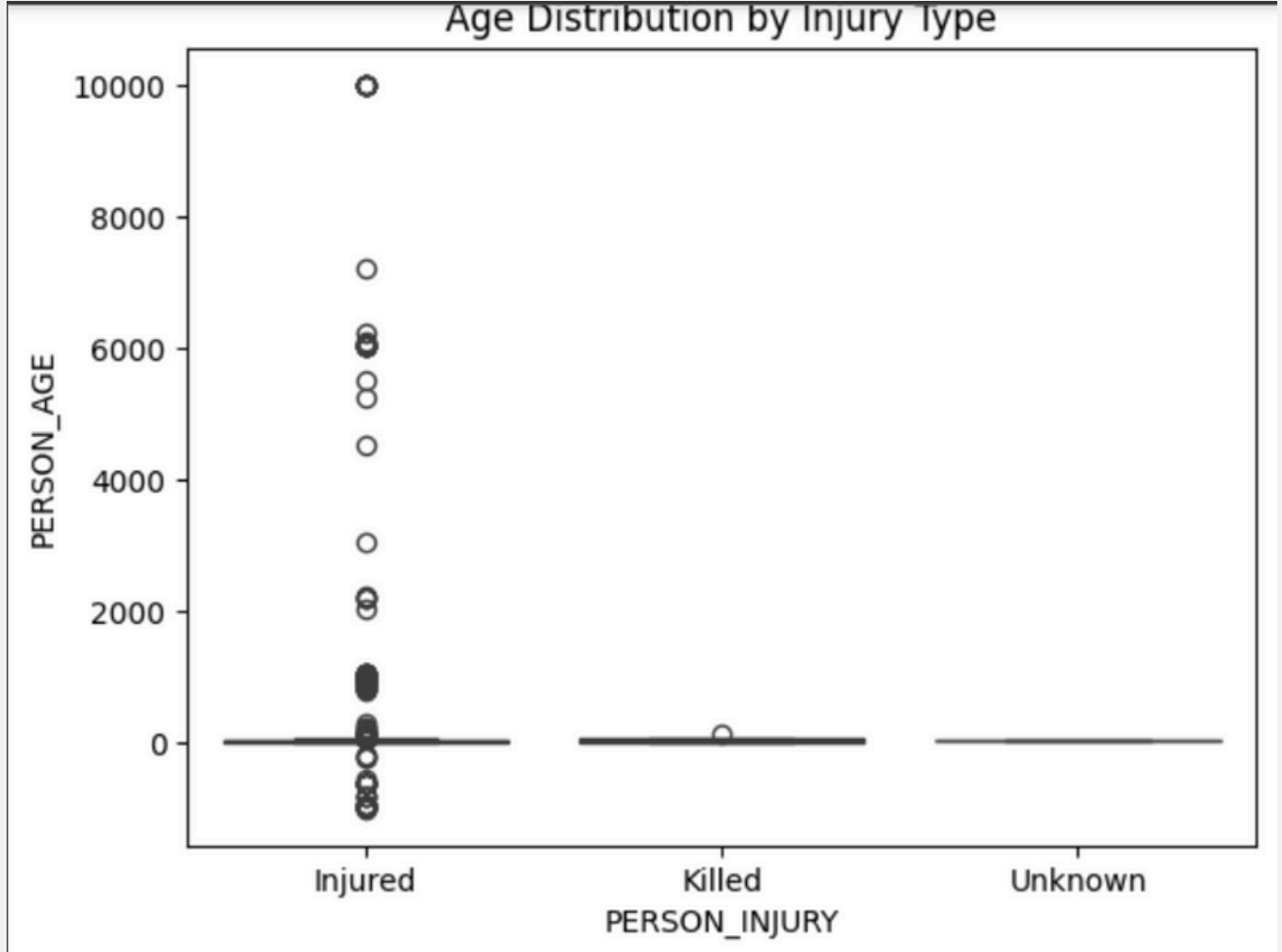
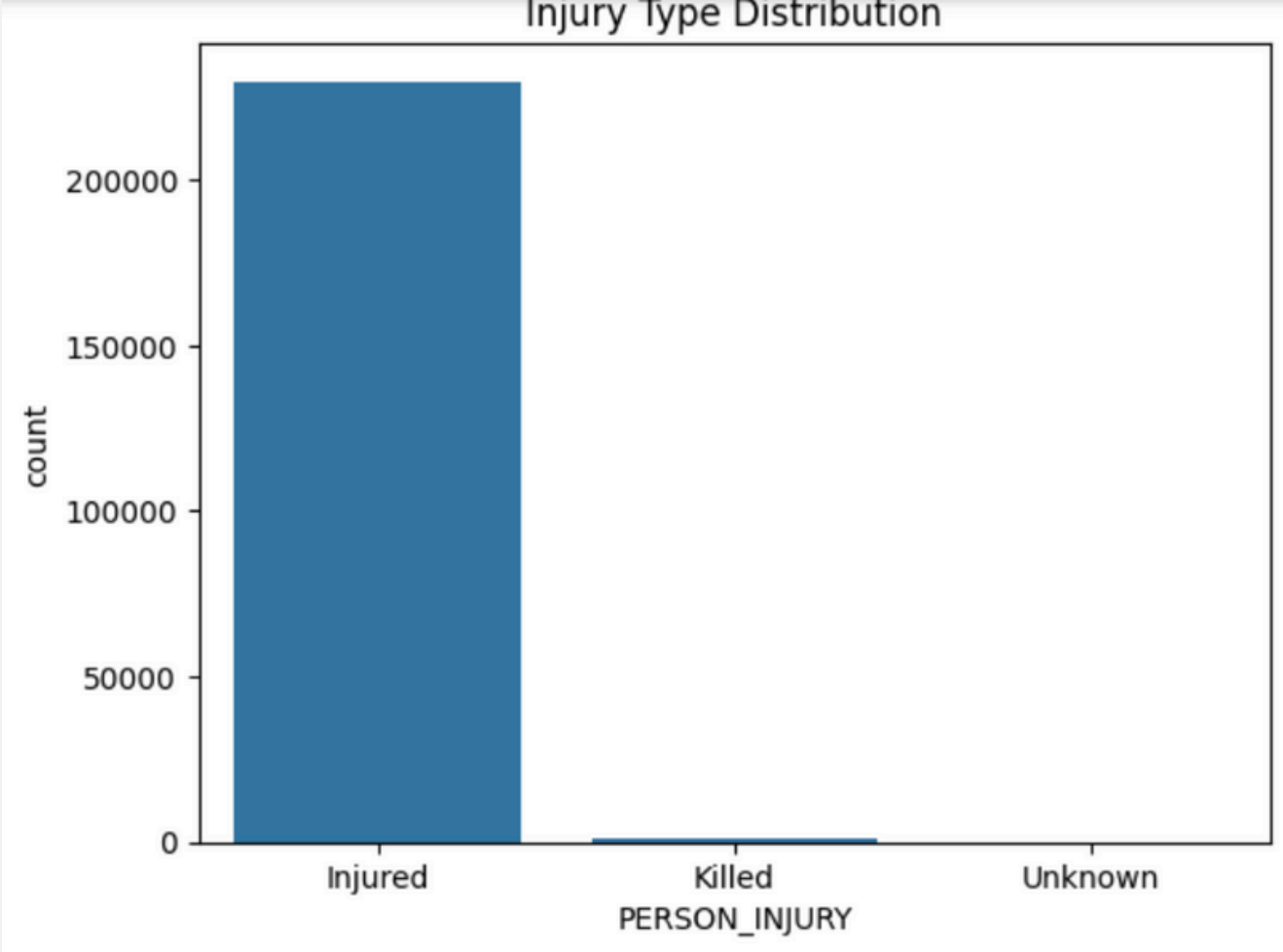
```
df.drop(['UNIQUE_ID', 'COLLISION_ID', 'CRASH_DATE', 'CRASH_TIME', 'PERSON_ID', 'VEHICLE_ID'], axis=1, inplace=True)

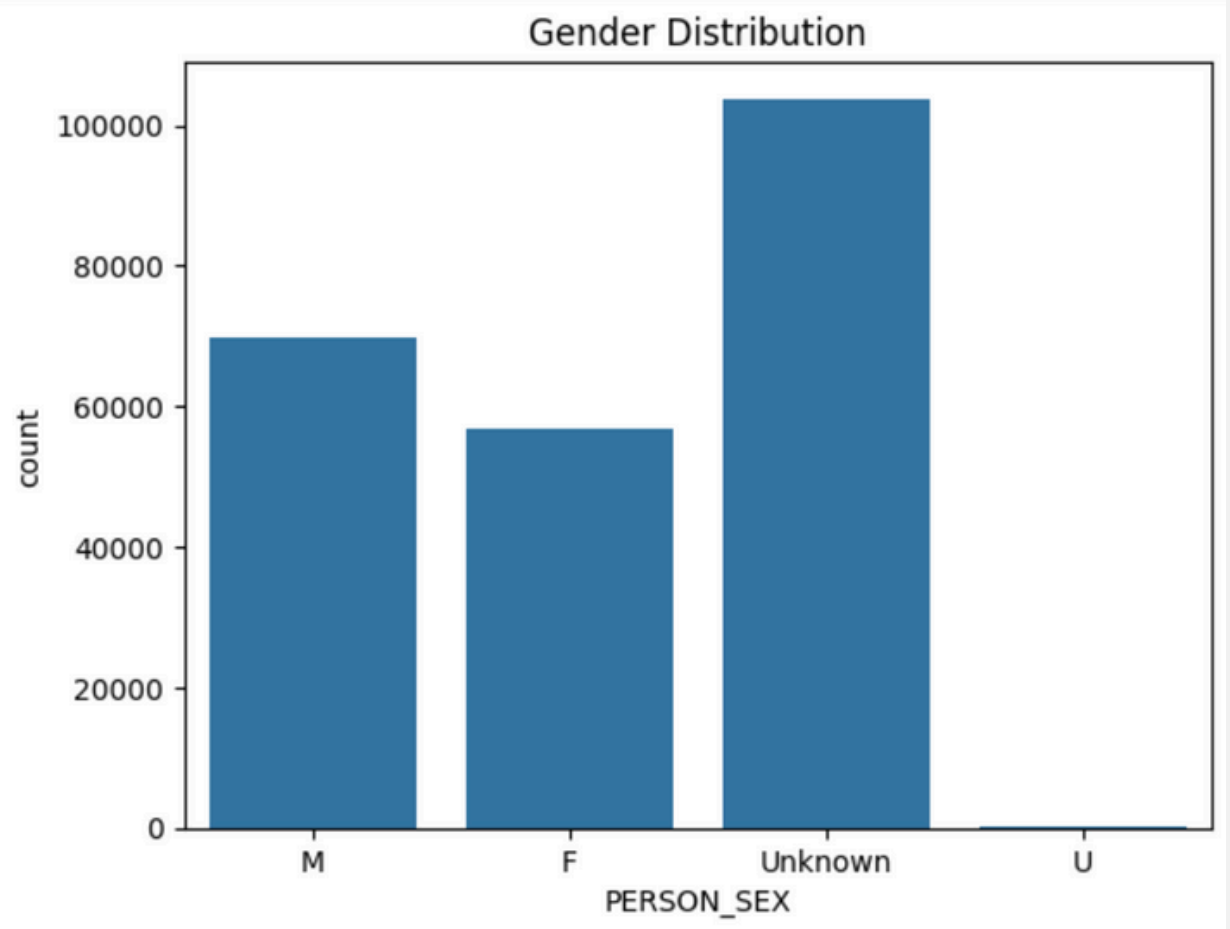
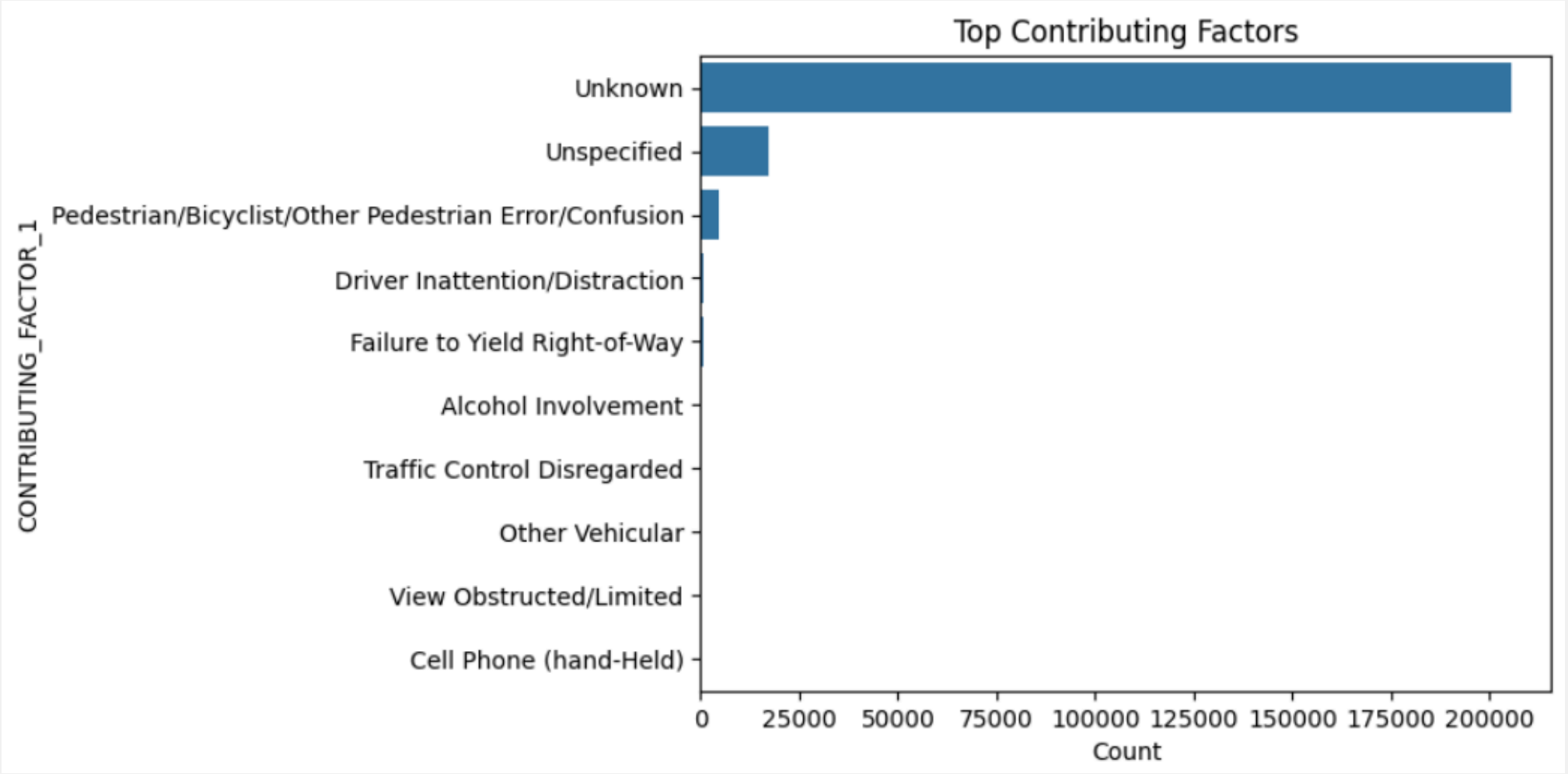
df['PERSON_AGE'] = pd.to_numeric(df['PERSON_AGE'], errors='coerce')

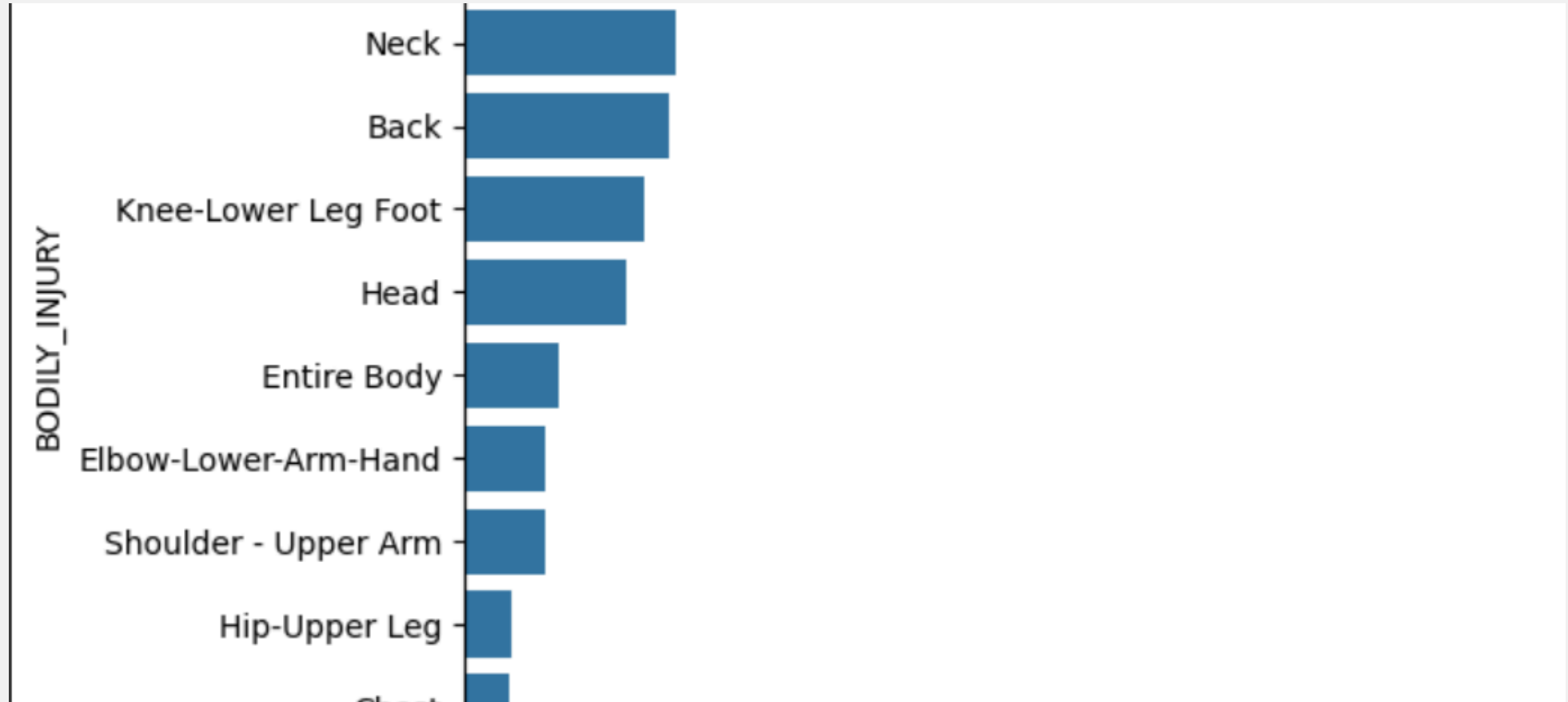
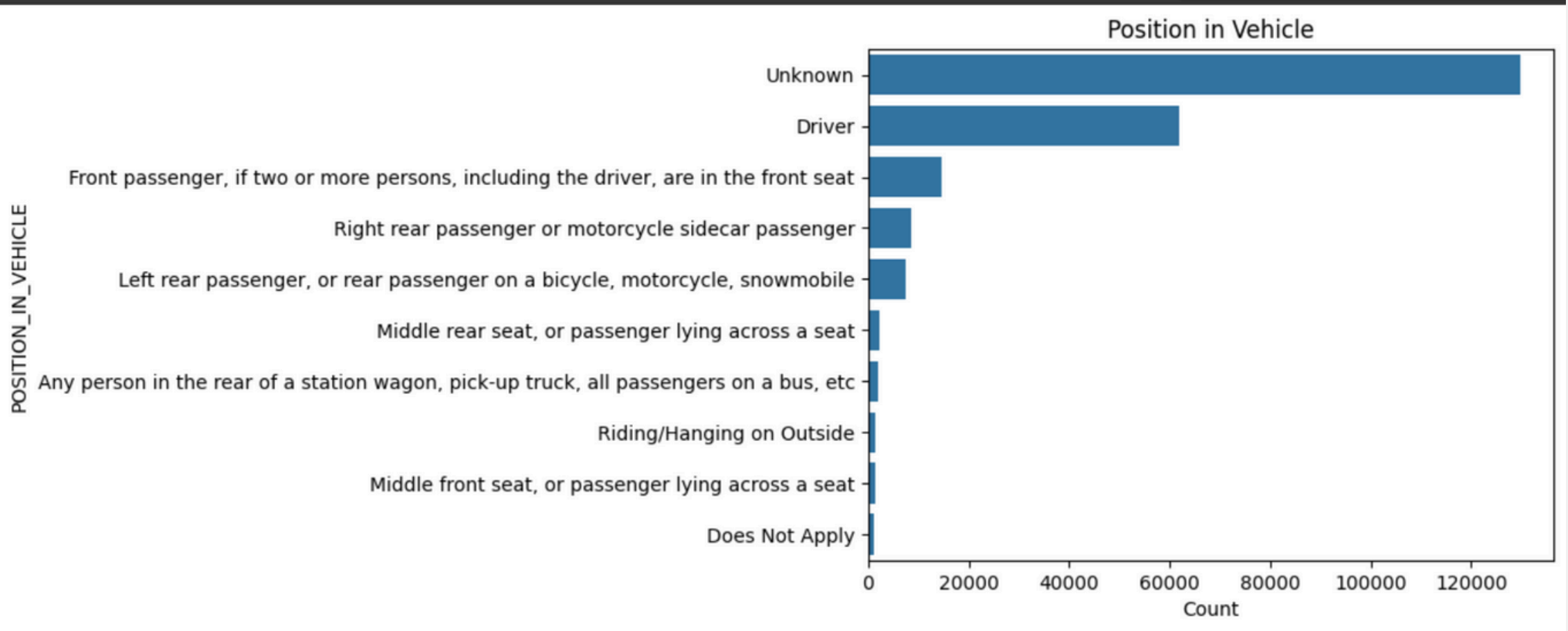
df['PERSON_AGE'].fillna(df['PERSON_AGE'].mean(), inplace=True)

df.fillna('Unknown', inplace=True)
```

EDA TO WORK ON KEY DETAILS







Future Improvements

- Incorporate real-time traffic and weather data
- Expand dataset across more regions
- Build a dashboard for real-time predictions
- Add multi-class severity levels (minor, serious, fatal)

Project Summary

This project aims to develop a machine learning model that predicts the severity of car accidents using real-world collision data. The goal is to support emergency responders with timely information, improve resource allocation, and identify key factors contributing to severe accidents. The model's performance insights are provided to guide safety improvements and future research.

Thank you very much!